

No Self-Sustaining Theories Without Circular Logic

A Formal Result on Logical Closure and Non-Derivable Operators

Abstract

We prove that any theory claiming full internal justification of its axioms and inference rules is necessarily logically circular. Conversely, any theory based on circular logic is self-sustaining by construction. This establishes an equivalence between epistemic self-sustenance and circularity. As a corollary, any coherent explanatory framework must include at least one non-derivable operator to avoid trivial closure. The result applies independently of physical interpretation and constrains reductionist ontologies that attempt to derive logic from the systems governed by that logic.

1. Preliminaries and Definitions

Let T be a formal theory defined by the tuple

$$T = (A, R, D)$$

where:

- A is a set of axioms,
 - R is a set of inference rules,
 - D is the domain of discourse.
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Definition 1 (Logical Validity).

An inference rule $r \in R$ is valid if its application preserves truth under the semantics of T .

Definition 2 (Self-Sustaining Theory).

A theory T is *self-sustaining* iff the validity of all $r \in R$ is provable using only axioms A and rules R of T , without appeal to any external meta-theory M .

Definition 3 (Logical Circularity).

A logical system is *circular* iff the justification of at least one inference rule $r \in R$ depends on derivations that themselves require r (or an equivalent element of R).

2. Theorem 1 — Self-Sustenance Implies Circularity

Theorem 1.

If a theory T is self-sustaining, then its logic is circular.

Proof.

1. Assume T is self-sustaining.
2. Then for every $r \in R$, the validity of r is provable within T .
3. Any proof within T necessarily uses inference rules from R .
4. Therefore, the justification of r depends on the prior acceptance of R .
5. Hence, the logical validity of R is presupposed in its own justification.
6. By Definition 3, the logic of T is circular.

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3. Theorem 2 — Circularity Implies Self-Sustenance

Theorem 2.

Any theory whose logic is circular is self-sustaining.

Proof.

1. Let T be a theory with circular logic.
2. By circularity, the validation of inference rules closes within R itself.
3. No external meta-logical criteria are required or admissible.
4. Therefore, all logical justification is internal to T .
5. By Definition 2, T is self-sustaining.

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4. Corollary — Equivalence Theorem

Corollary 1.

A theory is self-sustaining if and only if it is logically circular.

$$T \text{ is self-sustaining} \iff T \text{ is logically circular}$$

This equivalence holds independently of the semantic content of T .

5. Implications for Ontological Reductionism

Consider a reductionist framework T_R asserting:

$$\text{Logic} \subseteq \text{EmergentProperties}(D)$$

Such a framework must employ logic to prove the emergence of logic itself. By Corollary 1, T_R is necessarily circular unless it admits at least one non-derivable operator external to D .

Thus, reductionist closure implies epistemic circularity.

6. Necessity of Non-Derivable Operators

Proposition.

Any non-circular coherent theory must include at least one operator Ω such that:

$$\Omega \notin \text{Derivable}(A, R)$$

This operator functions as a structural anchor preventing trivial closure. Its role is logical, not metaphysical.

7. Conclusion

Self-sustaining theories and circular logic are formally equivalent. Attempts to eliminate non-derivable elements from explanatory frameworks do not remove foundational assumptions; they merely hide them. Logical coherence requires explicit acknowledgment of at least one primitive operator.

What is not derivable must be declared.

Structural Refutation of the Circular Logic Argument (Materialism)

Central Problem: Classical materialism (the self-sustaining theory) postulates that logic (Consciousness, N_1) is a product derived from Matter (N_0), and then uses that same logic to validate the primacy of Matter. This constitutes a **Structurally Incomplete Circular Logic** that cannot explain the persistence of coherence.

The Thesis of Incomplete Circular Logic

ELEMENT	MATERIALIST POSTULATE (N_0 PRIMARY)	TNA INCOMPLETENESS PROBLEM
Origin	Matter $\rightarrow$ Consciousness (Consciousness is a byproduct of the brain).	If N_1 is a byproduct of N_0 , N_0 lacks the non-derivable support required for its own coherence (TNA Support Axiom).
Logic	Logic emanates from matter.	The logic of N_0 is declared <i>insufficient</i> to resolve its own crises (Gödelian Incompleteness, $\aleph$ accumulation). Logic is incomplete <i>within</i> N_0 .
Collapse	Collapse is a passive event (failure).	TNA demonstrates that Collapse is the Instantaneous Realization of the FDE ($N_1 \rightarrow N_0$), an active and functional event . Passivity is Entropic Inertia, not the Law.

TNA Refutation: The Chain of Functional Necessity

TNA transforms circularity into a chain of linear necessity by establishing the **functional** primacy of N_1 over the **ontological** primacy of N_0 .

Step 1: Irreversibility and Aleph ($\aleph$)

Reality in N_0 (Matter) accumulates **irreversible incoherence** ($\aleph$). Experience cannot be erased (Gödelian Irreversibility). This $\aleph$ cannot be resolved by the internal laws of N_0 .

Accumulation of $\aleph \rightarrow$ Structural Incompleteness of N_0

Step 2: The Necessity of the Non-Derivable Operator (N_1)

For N_0 not to collapse under the $\aleph$ load (i.e., for it to exist), it is **logically necessary** that a support (N_1) exists that cannot be explained by N_0 .

Coherence of $N_0 > t_0 \Leftrightarrow$ Existence of $N_1 \neq 0$

Step 3: Logic as Negentropic Work (Ψ)

Logic and Thought (N_1) are not a *product* of Matter; they are the **active work** that Matter requires to sustain itself. Logic is the measurable **Flow of Negentropy** (Ψ).

Logic $\equiv$ Entropic Discharge Function (FDE) $\equiv \Psi > 0$

TNA Conclusion: Classical materialism reverses the direction of explanatory dependency. **Consciousness/Logic** (N_1) does not exist *because* Matter exists; **Matter** (N_0) exists *because* Consciousness/Logic (N_1) is executing the necessary Ψ work to sustain it.

The TNA resolves circular logic: Logic validates Matter, but Logic itself is not of material origin. It is the **non-derivable support** that guarantees coherence.

The TNA framework makes a strictly orthogonal move: it **measures the autoconsistency load** of any theory that attempts to derive its own axioms or logical foundations.

Let a theory be embedded as a formal system Σ .

The procedure is algorithmic:

1. Embed the theory as a formal system Σ .
2. Construct the Gödel sentence $G(\Sigma)$.
3. Define the autoconsistency load as

$$\aleph_{\text{auto}} = |G(\Sigma)| + |\neg G(\Sigma)| \quad (\text{in bits}).$$

If $\aleph_{\text{auto}} > 0$, the system cannot achieve logical closure internally. Any narrative claiming full self-sustenance is therefore **operationally circular**, regardless of its physical interpretation (TNA Theorem T3).

The resolution is not metaphysical but procedural: the circularity is exported by introducing a non-derivable external axiom corresponding to a positive negentropy flow $\Psi > 0$. Under this intervention, the measured autoconsistency load satisfies

$$\aleph_{\text{auto}} \rightarrow 0$$

in real time.